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Abstract—Graph embedding techniques are important tools for representing graphs in low-dimensional vector spaces. This study evaluates three methods, Graph2Vec, NetLSD and Graph Isomorphism Networks (GIN), on benchmark datasets IMDB-Multi, ENZYMES and MUTAG. Performance is assessed through graph classification and clustering tasks, considering predictive accuracy, computational efficiency and scalability. Results indicate that GIN achieves ??? the highest ??? classification accuracy but requires substantial computational resources. Graph2Vec balances performance and efficiency, while NetLSD provides competitive clustering performance with minimal overhead. Therefore it is apparent that each method has its distinct trade-offs. All methods maintain?? reasonable robustness under moderate perturbations, with sensitivity depending on dataset characteristics. These findings provide practical insights that aid in selecting graph embedding approaches based on the task requirements, dataset properties and computational constraints.

Index Terms—Graph Embedding, Graph2Vec, NetLSD, GIN, Graph Classification, Graph Clustering

I. INTRODUCTION

Chatia alla mou arese. Graphs are widely used to model complex relational data in domains such as social networks, bioinformatics, and chemistry. Applying standard machine learning methods to graphs is challenging due to their irregular structure. Graph embedding techniques address this by mapping graphs into low-dimensional vector spaces that preserve structural information, enabling tasks such as classification and clustering.

Several embedding methods exist with different strengths. Graph2Vec captures subgraph patterns to produce graph-level embeddings efficiently. NetLSD encodes graphs via spectral descriptors, providing compact and calable representations. Graph Isomorphism Networks (GIN) use neural architectures to aggregate

neighborhood information, achieving high predictive performance but at higher computational cost.

In this study, we evaluate these three methods on IMDB-MULTI, ENZYMES and MUTAG, comparing accuracy, robustness, scalability, and computational efficiency. The results highlight the trade-offs of each method and provide guidance for selecting appropriate embeddings based on dataset characteristics and task requirements.

II. RELATED WORK

Briefly discuss prior graph embedding techniques (unsupervised vs. supervised).

Mention previous benchmarks and known trade-offs.

III. METHODS

Graph2Vec: how it captures subgraph patterns.

NetLSD: spectral descriptors, computational efficiency.

GIN: neighborhood aggregation, supervised learning approach.

Implementation details: libraries used (KarateClub, PyTorch Geometric), embedding dimensionality, training setup

IV. DATASETS

Describe MUTAG, ENZYMES, IMDB-MULTI: type of graphs, nodes, labels.

Any preprocessing steps.

V. EXPERIMENTS AND EVALUATION

A. Classification

SVM/MLP classifier for embeddings.

Report accuracy, F1-score, AUC.

Computational metrics: training time, generation time, memory usage.

Analysis of embedding dimensionality vs. performance.

B. Clustering

k-means/spectral clustering.

ARI scores and qualitative visualization (t-SNE or UMAP).

Discussion of cluster separation quality.

Identify applicable funding agency here. If none, delete this.

C. Stability Analysis

Random perturbations (edges/nodes).

Compare embedding changes and drop in classification accuracy.

VI. RESULTS

Tables and figures summarizing all tasks.

Highlight trade-offs: performance vs. efficiency vs. stability.

REFERENCES

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